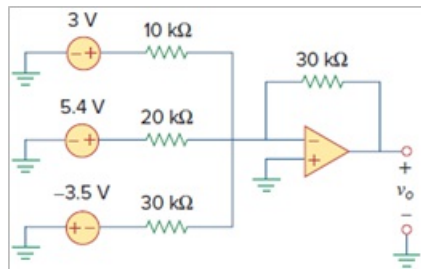


Problem

Determine the output of the summing amplifier in Fig. 5.74.

Figure 5.74:

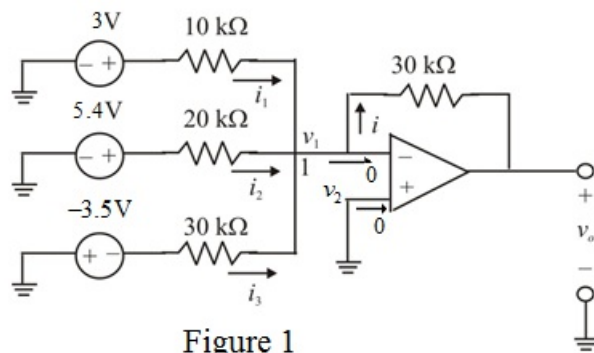


Step-by-step solution

Step 1 of 3

Refer to Figure 5.74 in the textbook.

Let v_1 be voltage the inverting terminal and v_2 be the voltage at the non-inverting terminal of op-amp as shown in Figure 1.



Step 2 of 3

In an ideal op-amp, the inverting and non-inverting terminal currents are zero.

The inverting and non-inverting node voltages of an op-amp are equal.

$$v_1 = v_2$$

The non-inverting terminal of an op-amp is grounded.

$$v_2 = 0 \text{ V}$$

Therefore, $v_1 = v_2 = 0 \text{ V}$.

Use nodal analysis at node 1 as shown in Figure 1.

$$i = i_1 + i_2 + i_3$$

$$\frac{v_1 - v_o}{30 \times 10^3} = \frac{3 - v_1}{10 \times 10^3} + \frac{5.4 - v_1}{20 \times 10^3} + \frac{3.5 - v_1}{30 \times 10^3}$$

$$\frac{v_1 - v_o}{3} = \frac{3 - v_1}{1} + \frac{5.4 - v_1}{2} + \frac{3.5 - v_1}{3}$$

$$\frac{v_1 - v_o}{3} = \frac{18 - 6v_1 + 16.2 - 3v_1 + 7 - 2v_1}{6}$$

$$2v_1 - 2v_o = 41.2 - 11v_1$$

$$13v_1 - 2v_o = 41.2 \dots\dots (1)$$

Step 3 of 3

Substitute 0 for v_1 in equation (1) to obtain v_o .

$$13v_1 - 2v_o = 41.2$$

$$13(0) - 2v_o = 41.2$$

$$v_o = -20.6 \text{ V}$$

Therefore, the output of the summing amplifier is -20.6V.

NOTE: If we consider the magnitude of voltage source as 3.5 rather than -3.5, then we get the text book answer as -13.6V.

[Comments \(5\)](#)

☐

Anonymous

use the equation for summing amplifiers, $V_o = [(R_f V_1)/R_1 + (R_f V_2)/R_2 + (R_f V_3)/R_3]$

☐

Anonymous

Answer is $V_o = 7 \text{ V}$ (I believe)

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Anonymous

or -7 V since inverting

☐

Anonymous

The answer by the book is - 13.6 V

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Anonymous

_13.6 is the correct answer by using the equation of summing amplifiers